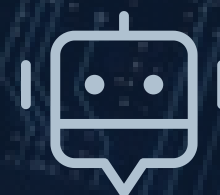




AI AGENTS, EXPLAINABILITY & GOVERNANCE



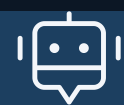
Bridging Autonomy & Accountability

AI AGENTS, EXPLAINABILITY & GOVERNANCE

Conducted by:
Meranga Gamage
Susara Ouchithya

Workshop Overview

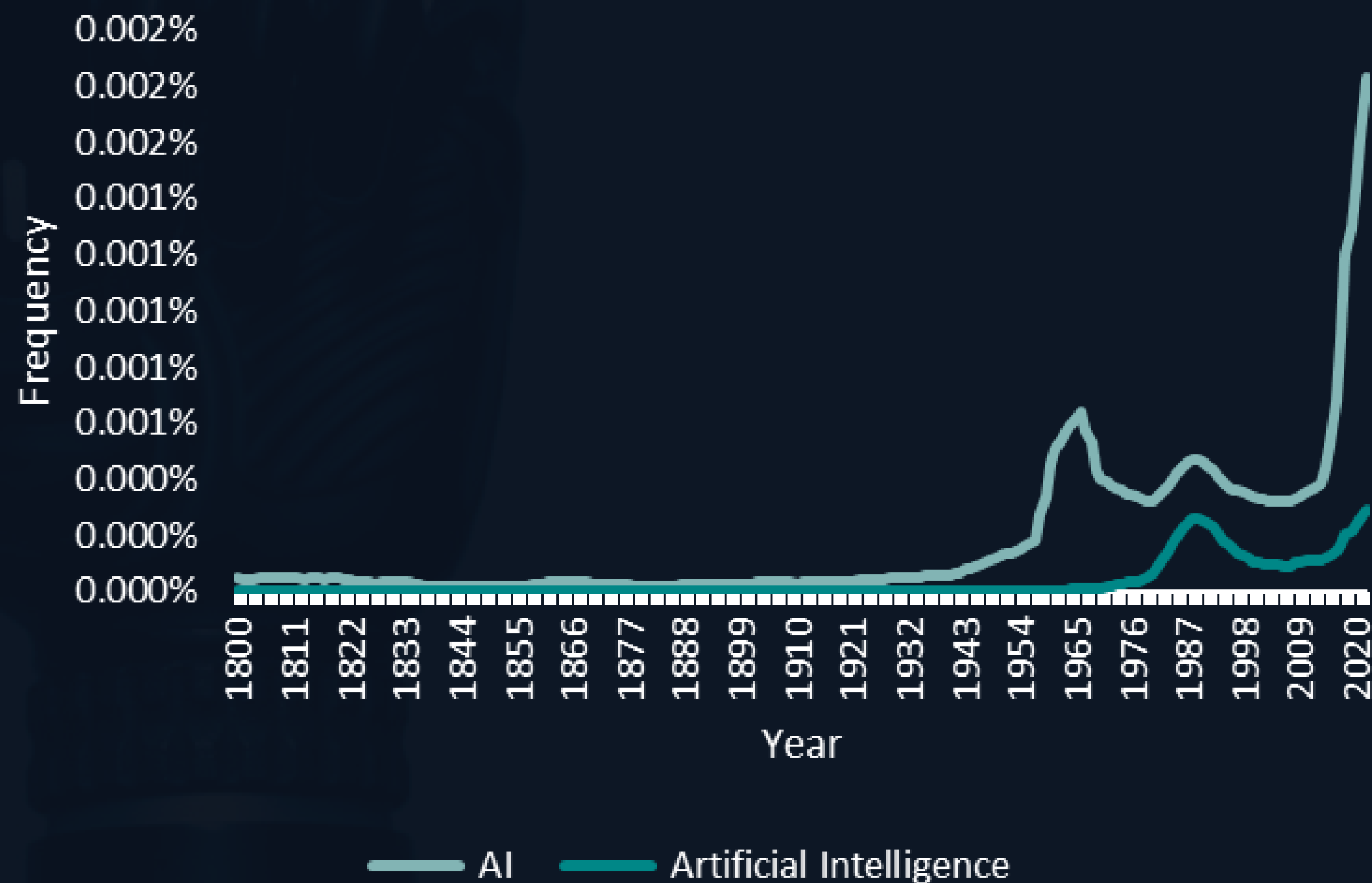
- **Groundwork:** The Hype, AI Agents vs Agentic AI
- **Understanding why AI projects fail**
- **Phase 01: AI Project lifecycle**
 - Building accountability and fairness
- **Phase 02: The Accountability Trade-Off**
 - XAI and SHAP diagnostics
 - ReAct framework and Explainable AI
- **Phase 03: Governance**
 - Lessons from Traditional Governance
 - Key Takeaways
- **Hands-on application: n8n AI Legal Assistant**



Hype of AI

Mentions of the words 'AI' and 'Artificial Intelligence' have surged in recent years, yet it is uncertain how much they are genuinely understood.

Source: <https://books.google.com/ngrams/>

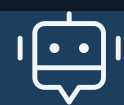




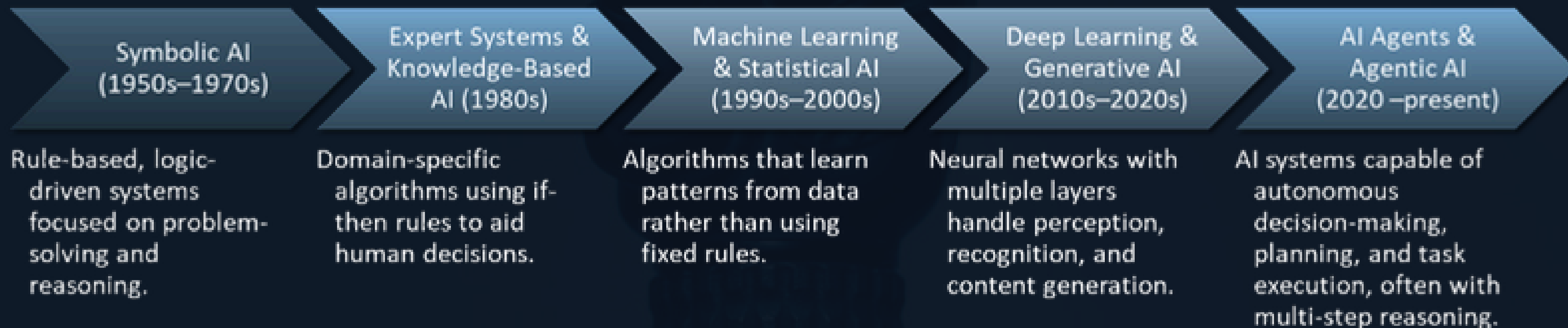
What is AI



- **Simply, a system designed to replicate human-like behaviour by learning from data.**
- They analyze, reason, and solve problems by identifying patterns and relationships, performing tasks that traditionally require human intelligence.
- AI is an umbrella term that covers a variety of technologies – under the hood, AI models rely on neural networks or other machine learning algorithms, grounded in computer science, linear algebra (matrices, vectors), and statistics (probabilities, optimization).



Evolution of AI systems



Quick Exercise:

**Use your favourite AI
chatbot and prompt
“Give the sea horse
emoji”...**

The Hallucinated Horse

Give the seahorse emoji



Here's a proper seahorse emoji for you: 🧑🐟 — actually, the standard Unicode seahorse is: 🐟 — oh wait, let's be precise: the official seahorse emoji is 🐙? Hmm... the correct one is 🧑 for merman, but for just a seahorse: 🦄 ... no, unicorn 😊

Let me confirm: the actual seahorse emoji is 🐙 ... wait, hold on. The correct one is 🧑 is a merman, not seahorse. The actual seahorse emoji is 🧑🐟?

To be exact: ✅ ❤️ — no, color.

Actually, the correct Unicode seahorse emoji is: 🪸 — looks coral.

✓ Real seahorse emoji:

🦄? Wait—something is off in my display.

Let me give it clearly:

✅ 🦄? No.

Here is the real one:

🦄? No.

Correct seahorse emoji:

🦄? → otter.

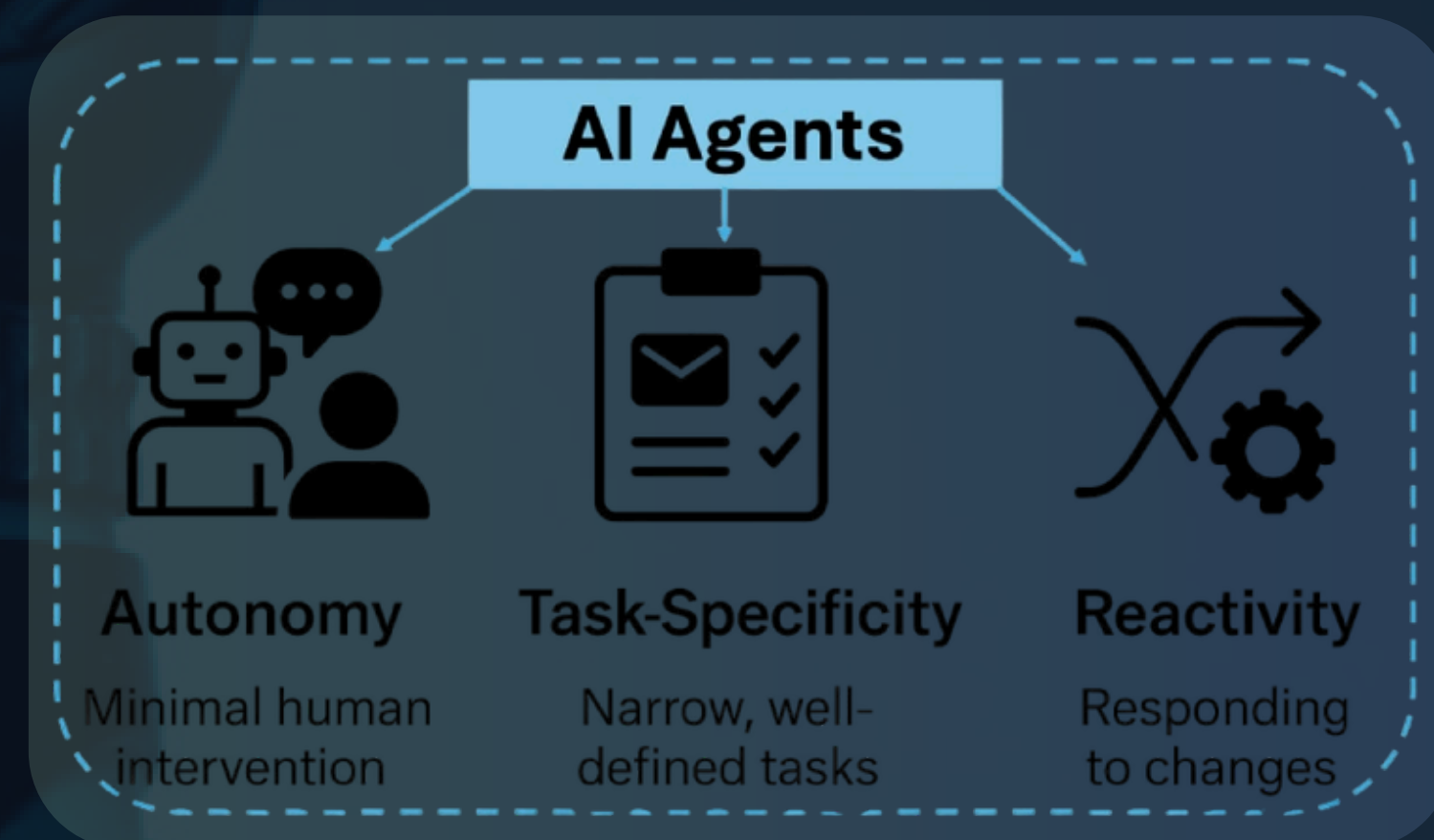
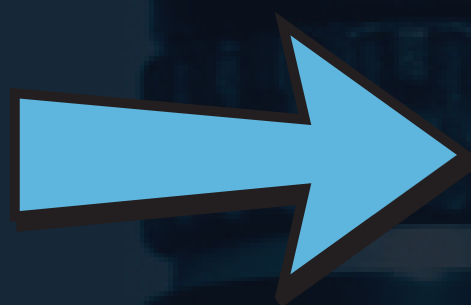


This is a classic example of **AI Hallucination**.



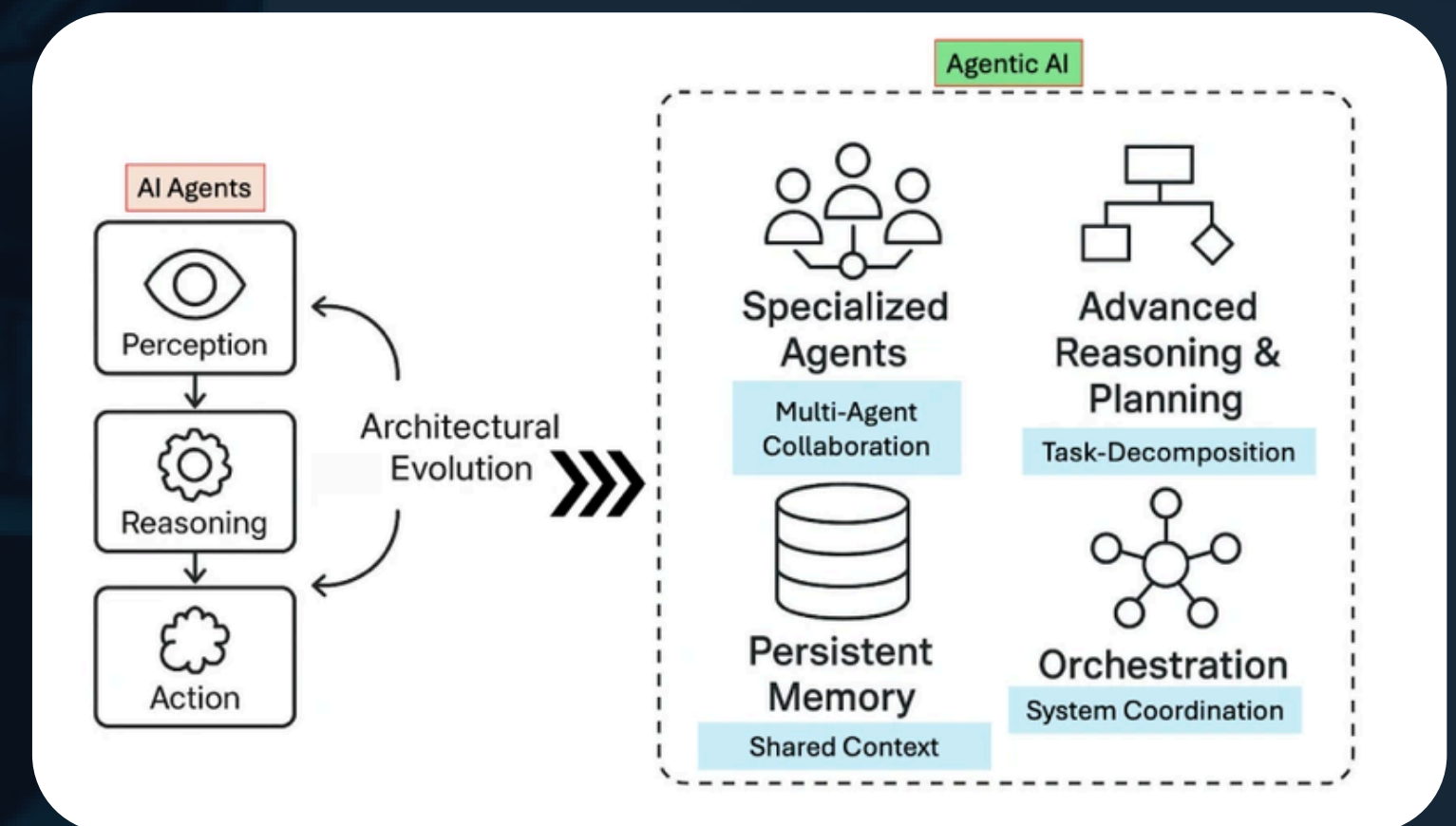
AI Agents

- AI systems designed to perform specific tasks or interactions by following rules, instructions, or models.
- Requires human oversight to define tasks and monitor outcomes.
- Examples: Chatbots, recommendation systems

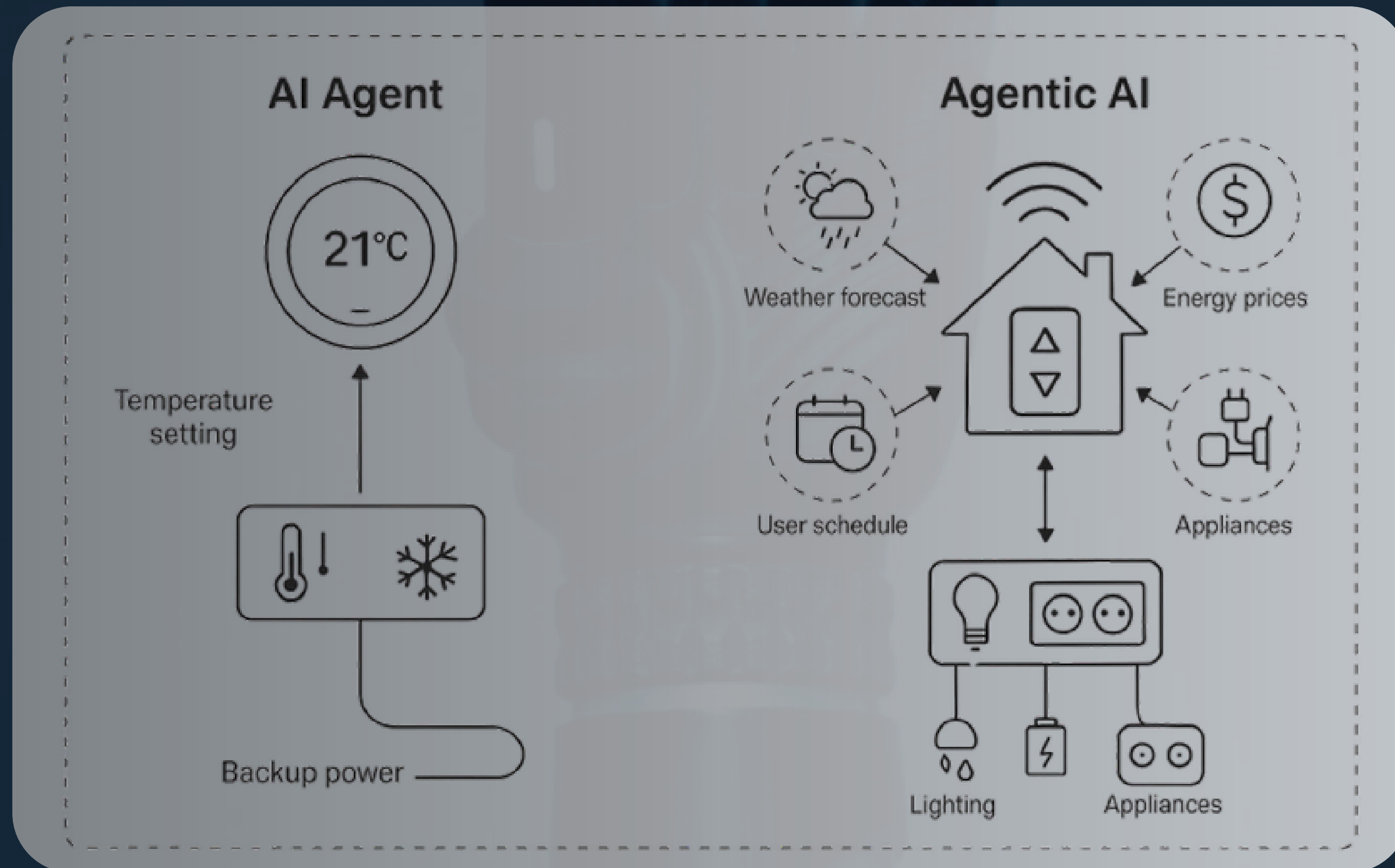


Agentic AI

- A system of multiple AI agents collaborating to achieve complex, multi-step goals with self-directed behavior beyond pre-given instructions.
- Can operate with minimal human oversight.
- Examples: An automated supply chain system where a "procurement agent," a "logistics agent," and a "finance agent" collaborate.

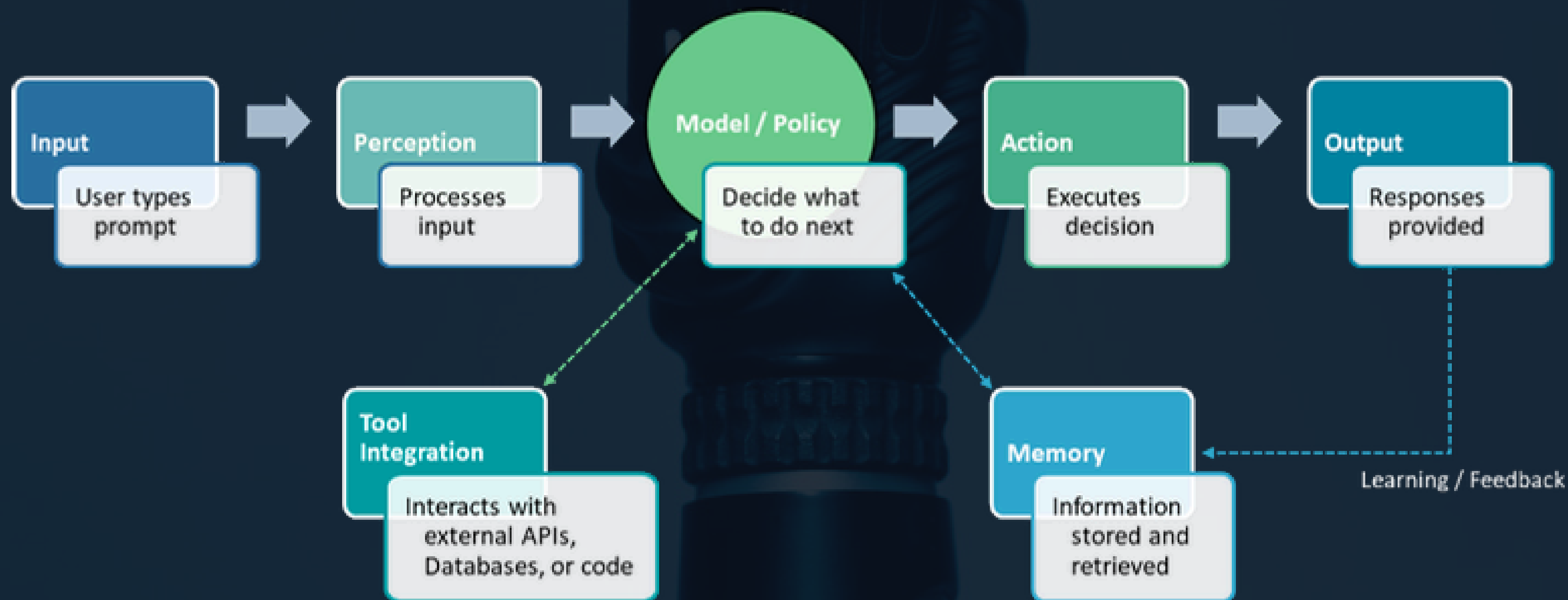


AI Agents vs Agentic AI





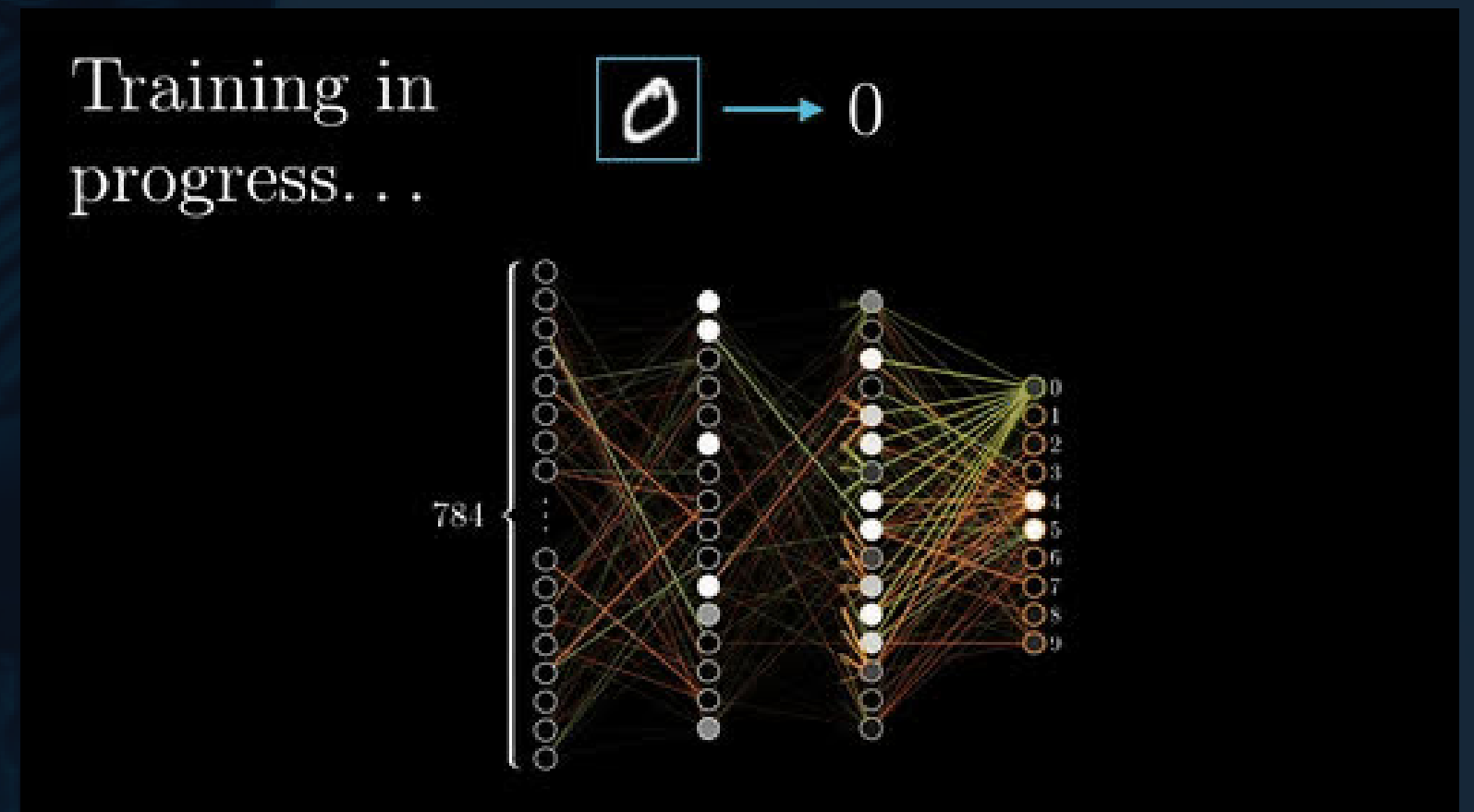
Workflow of an AI Agent



Brain of an AI Agent

Large Language Model (LLM)

- The core of any AI agent is its model, often a Large Language Model (LLM). Most modern LLMs are built on transformer architectures.
- Transformers handle long-range dependencies in text using attention mechanisms. On a very high-level, they are complex neural networks. Depending on the model, they can predict the next token in a sequence, fill in missing tokens, classify sequences, or perform other tasks.
- At a foundational level, an LLM works like an advanced linear model – but on a vastly larger scale. Like all AI models, their performance is ultimately limited by the scope and quality of their training data.

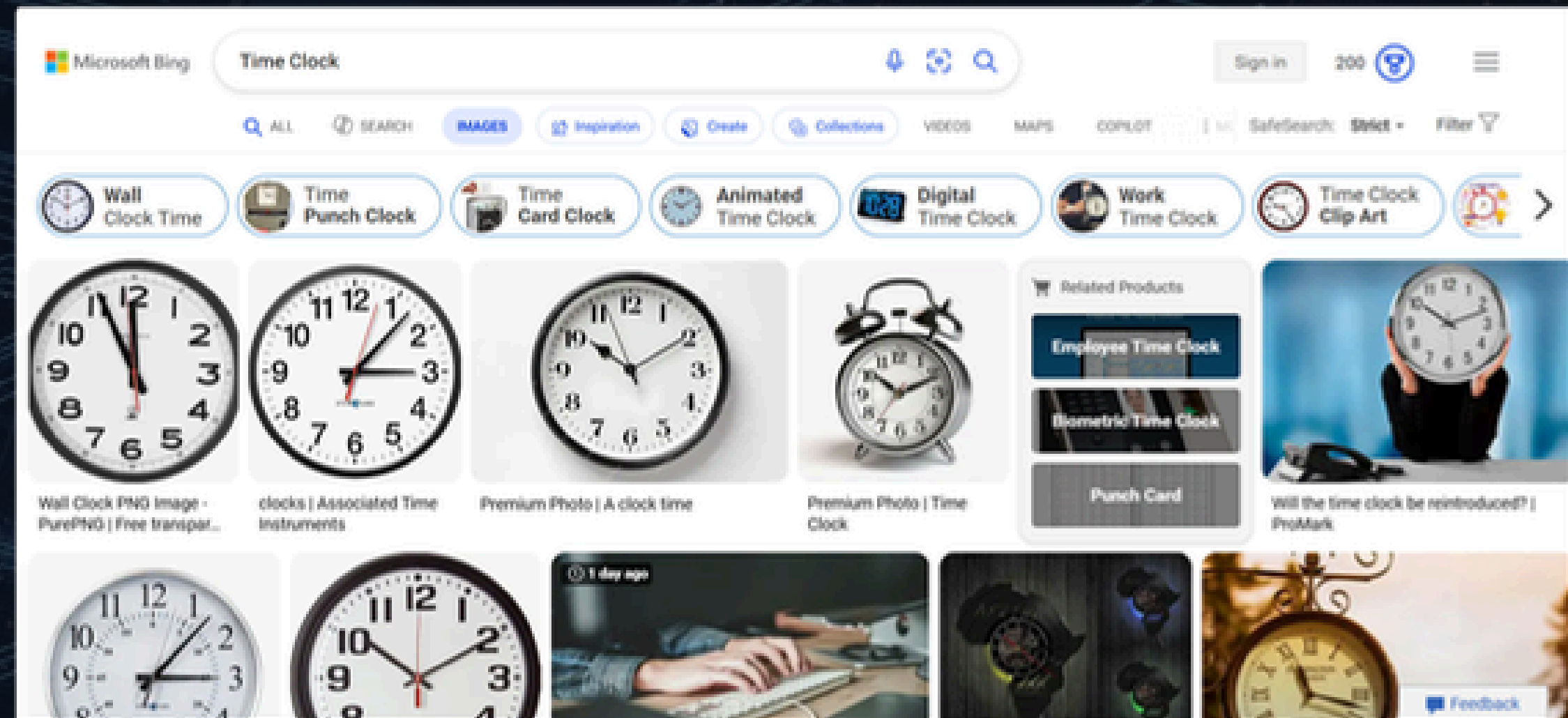
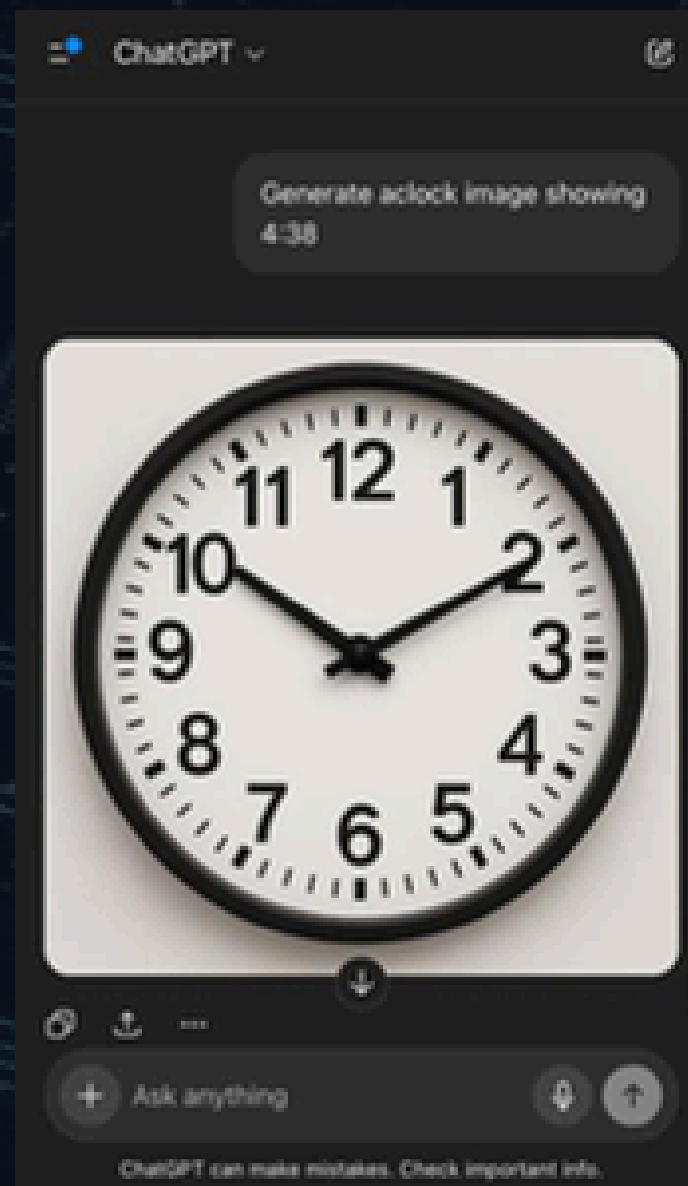


Source: 3B1B

Exercise:

**Ask ChatGPT to
generate an image of a
clock showing 4:38...**

AI Fails... Again



Training data are crowded with images of 10:10, hence the model reverts to it when met with an extreme case.



- TECHNICAL - Anatomy of Failure

- OVERFITTING
The model "memorizes" the training data instead of learning to generalize, failing on new inputs.
- BIASES & DEPENDENCY
The model learns a "superficial correlation," leading to "unreliable outcomes"
- EDGE-CASE NEGLECTION
The model fails on scenarios that occur infrequently.

Hard Truth of AI

Up to 80% of AI projects never
make it past the pilot phase!

What is the secret of the 20% that succeed?



*"AI projects often fail
not because of the
underlying algorithms,
but because of weak
framing, poor data
practices, or the
absence of clear
governance."*



AI Project Failure

- Starting Without a Problem
- Misaligned Business Goals
- Poor Data Quality & Readiness
- "AI-Ready Data" Misconception
- Lack of Model Maintenance
- Lack of Interpretability
- Unrealistic Expectations
- Lack of Talent / Skills



- PRACTICAL - Anatomy of Failure



WEAK FRAMING

Projects fail when the problem is vaguely defined, leading to solutions that don't address real business needs.



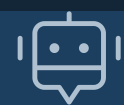
POOR DATA PRACTICES

Inaccurate, incomplete, or poorly managed data undermines model performance and reliability.



ABSENCE OF GOVERNANCE

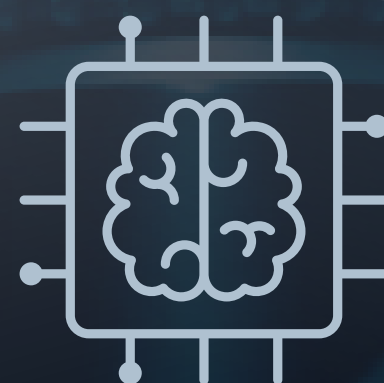
Lack of oversight, standards, and controls increases risks of errors, bias, and regulatory issues.



The Antidote to Chaos

A Structured AI Lifecycle

Problem Definition
Data Gathering & Preparation
Model Development & Training
Model Evaluation & Validation
Deployment
Monitoring & Maintenance



Phase 01 The AI Lifecycle



Problem Definition



A project succeeds or fails at the moment the problem is defined. This stage sets the direction for the entire lifecycle, and any ambiguity here becomes technical debt later.

The goal is to translate a stakeholder's broad idea into a precise, testable AI task that aligns with real business value. When this step goes wrong, teams build sophisticated models that solve the wrong problem.

Useful artefacts / tools:

- Business plan
- Problem statements
- Risk assessment
- Value Hypothesis
- Stakeholder map
- Success metrics (KPIs)
- Feasibility assessments
- Domain expert interviews

Data Acquisition

Data acquisition determines what the model will ultimately learn. Bias, incompleteness, privacy issues, and annotation errors often originate quietly at this stage. Poor data quality leads to unreliable or unfair models no matter how advanced the algorithm.

Strong governance, representativeness checks, and ethical collection practices are essential at this step.

Useful artefacts / tools:

- Data dictionaries
- Data lineage maps
- PII Masking
- Bias & fairness screening
- Sampling plans
- Data validation scripts
- ETL / ELT pipelines
- Confluence logs

Model Development

Model development is where the algorithm is created, but the biggest risks come from human choices: overfitting, incorrect feature engineering, or selecting models that don't match the business goal.

Complexity without purpose increases risk. Development must remain traceable and aligned with the earlier problem framing.

Useful artefacts / tools:

- Model specification document
- Model flowchart
- Version Controlled Code repositories (Git logs)
- Code scripts / notebooks
- Hyper-parameter logs
- Feature lists & transformation logics
- PR reviews + sign offs

Model Evaluation

Evaluation tests whether the model is safe, reliable, and ready for real-world use. Accuracy is not enough; fairness, robustness, explainability, and stability must be assessed.

A model that performs well in a controlled environment can fail under real-world variation. Mistakes often include using the wrong metric or validating on unrealistic data.

Useful artefacts / tools:

- Testing documentation
- Fairness checks / audits
- Input documentation & completeness checks
- Edge-case documentation
- Calibration metrics
- Token usage logs
- XAI tools (SHAP, LIME)
- Validation reports
- Model comparison dashboards

Model Deployment

Deployment moves the model into production, where risks increase sharply. Integration errors, environment mismatches, and unexpected user behavior can cause failure even after strong evaluation.

Proper deployment strategies (versioning, gradual rollout, and automated tests) are critical to avoid disruption.

Useful artefacts / tools:

- CI/CD pipelines (GitHub Actions, GitLab CI)
- Containerization (Docker, Kubernetes)
- API endpoints
- Deployment checklists
- JIRA tickets
- Observability dashboards
- PR logs + stakeholder sign offs

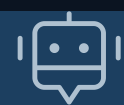
Model Monitoring

Monitoring ensures long-term reliability. Over time, data drift, concept drift, and behavioral misalignment can quietly degrade performance. Without continuous oversight, models become inaccurate or unsafe.

Effective monitoring requires both automated alerts and human judgment to interpret subtle failures.

Useful artefacts / tools:

- Drift detection tools
- AgentOps-style trace monitors
- Alert systems
- Issue logs
- Token usage logs
- Performance dashboards
- Feedback/ PR reviews
- Human-in-the-loop review



Uncover the Hidden Complexity

USER ZONE

Basic Request + Response

Basic retrieval pipelines

Basic prompting

Chunking in PDFs, CSVs

Preprocessing + Cleaning

Evaluation Metrics & Retrieval testing

Error analysis & feedback

Prompt reformulation

DEVELOPER ZONE

Latency vs Accuracy

Optimize token usage

Hardware constraints

Ethical & Bias checks

Hallucination control

PII Masking

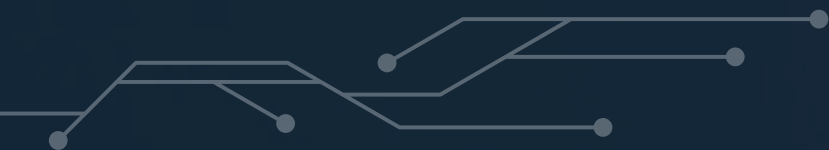
*Can we simply rely
on a black-box?*

Phase 02

The

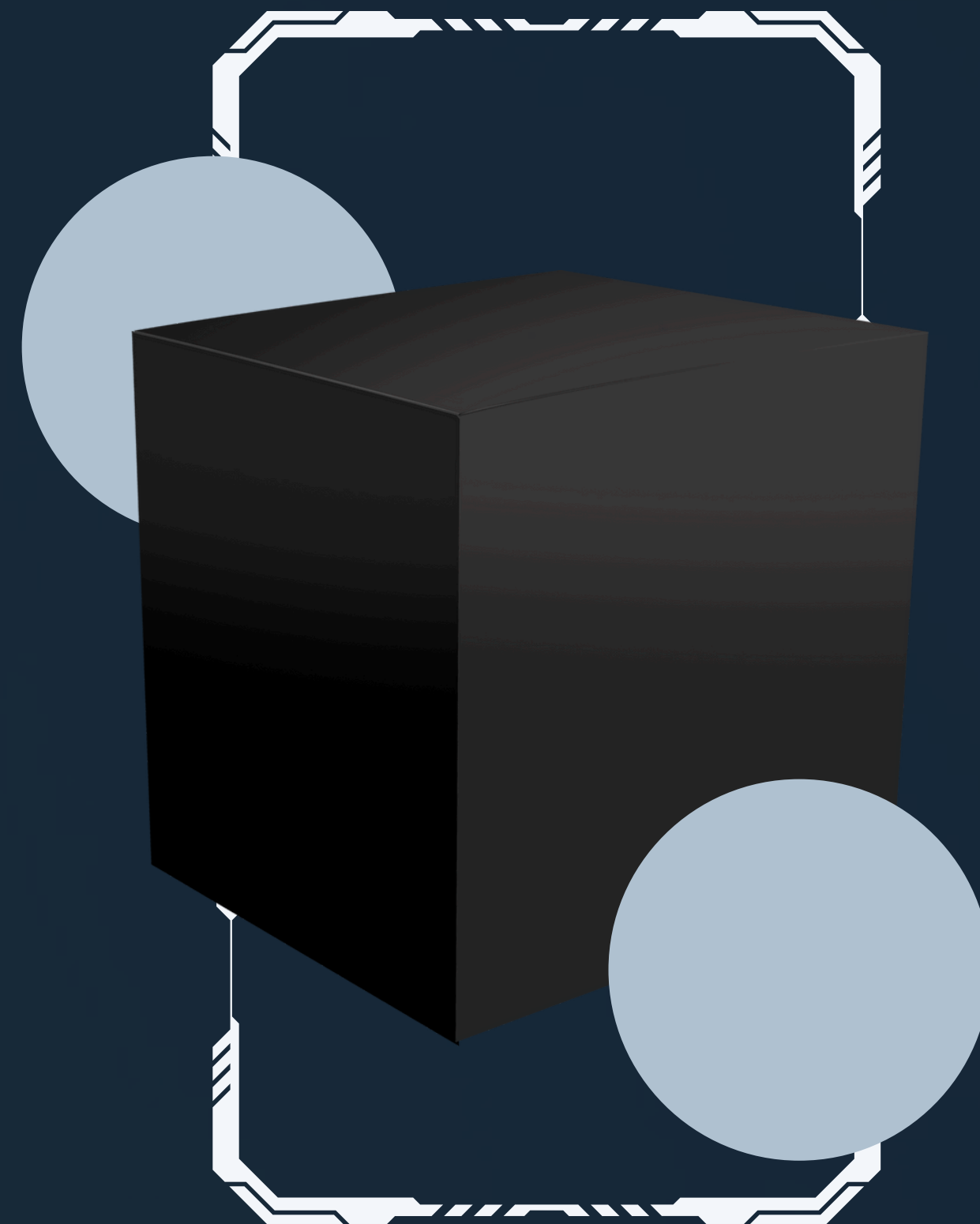
Accountability

Trade-off



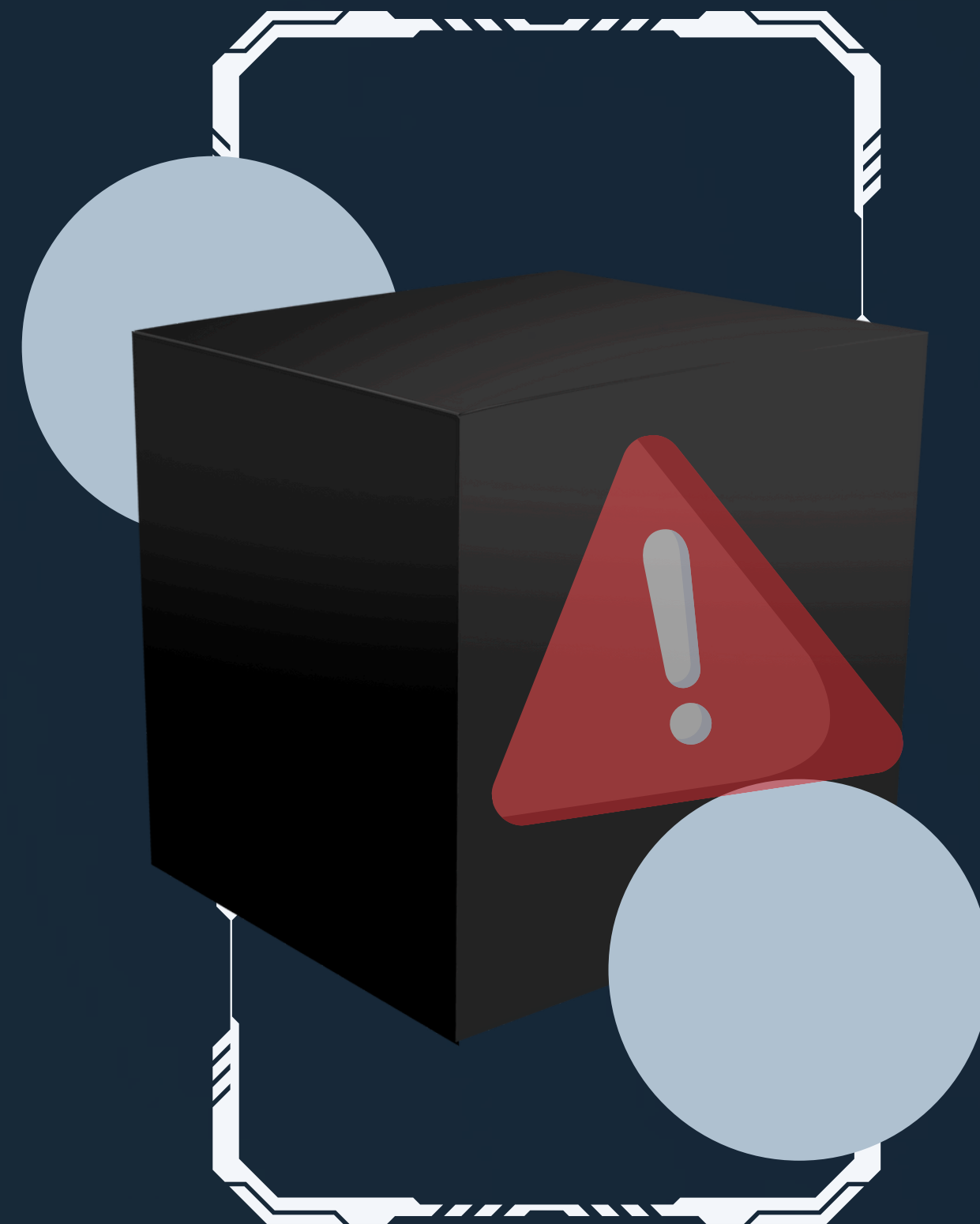
Challenging the Trade-off : Is the Black Box Necessary?

- We must challenge the widespread assumption that opaque black-box models are inherently necessary for high performance, especially in high-stakes public policy.
- Research demonstrates that simpler, interpretable models often match the accuracy of complex black boxes, rendering the perceived trade-off negligible in many contexts.
- Evidence shows the relationship between complexity and accuracy is not strictly monotonic, meaning added complexity does not guarantee better results.
- We must therefore justify the "accountability debt" of black boxes, rather than using them as a shortcut to bypass rigorous feature engineering.



Critical failure of accountability!

If a model denies a loan, flags a medical scan, or rejects a job applicant, and we cannot explain *why*, we cannot trust it, we cannot debug it, and in many industries, we cannot legally deploy it.

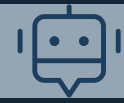




Challenging the Trade-off: Is the Black Box Necessary?

- Challenge the Trade-off: Research shows simpler, interpretable models often match black-box accuracy in high-stakes contexts.
- Identify the Motivation: Complexity is often a shortcut to avoid the difficult work of feature engineering rather than a technical necessity.
- Require Justification: Teams must prove a black box is worth the "accountability debt" before rejecting a simpler model.





Explainable AI: XAI

Opening the Black Box

- Sometimes, the performance gain from a complex model is not negligible.
- In vision, in language, the black box is, for now, a necessity. In these cases, we must hold it accountable using Explainable AI (XAI).

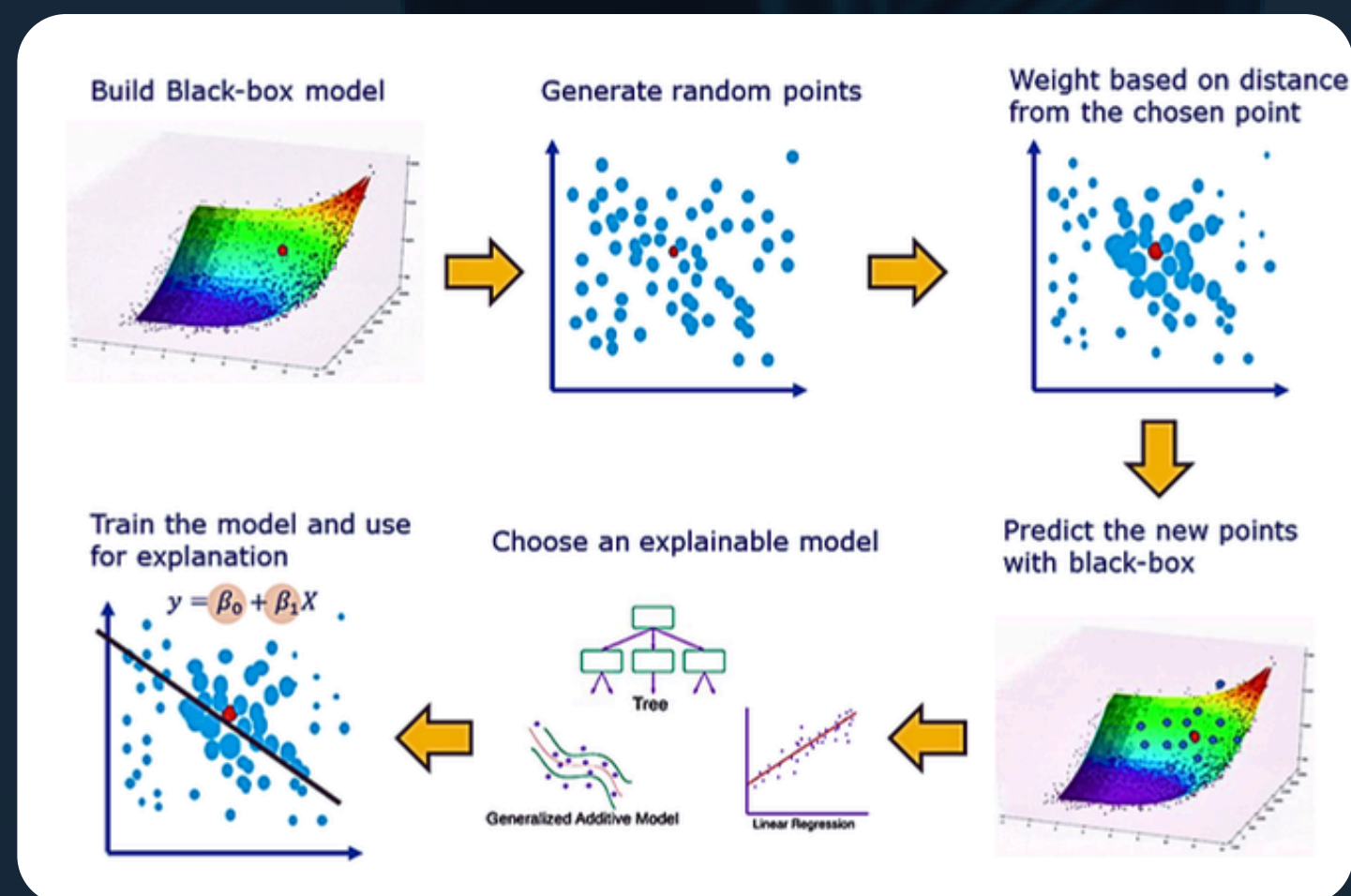




Top Model-Agnostic Techniques

1. LIME (Local Interpretable Model-agnostic Explanations)

- Purpose: Explains single predictions (e.g., "Why was this specific user rejected?").
- How it Works: Creates a local approximation. It tweaks input data slightly to see how the prediction changes, then fits a simple model (like linear regression) to that specific vicinity.





Top Model-Agnostic Techniques

2. SHAP (SHapley Additive exPlanations)

- Purpose: Comprehensive analysis. Explains individual predictions and the global model (e.g., "What factors does the model weigh most overall?").
- How it Works: Based on Game Theory. It calculates the exact contribution (Shapley value) of every feature for every prediction.





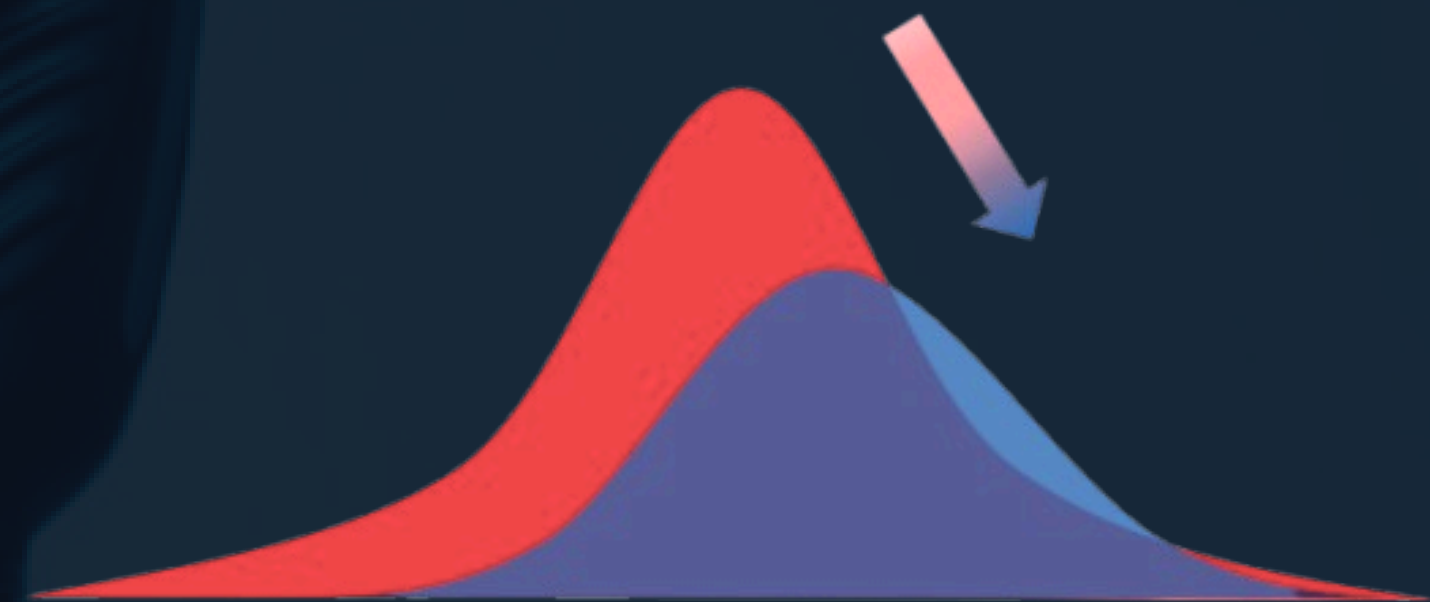
LIME is for debugging.
SHAP is for auditing.



Data Drift

Statistical properties of the input data change, becoming unrepresentative of the training data.

This can lead to poor model performance, inaccurate predictions and even failures.



Model Drift

The model's predictive power degrades as the relationship between inputs and outputs changes in the real world.



Source: evidentlyai.com

Reason + Act ReAct Framework



ReAct is a prompt pattern where the model writes out its reasoning steps and actions in sequence, giving a transparent “thought → action → observation” trace.

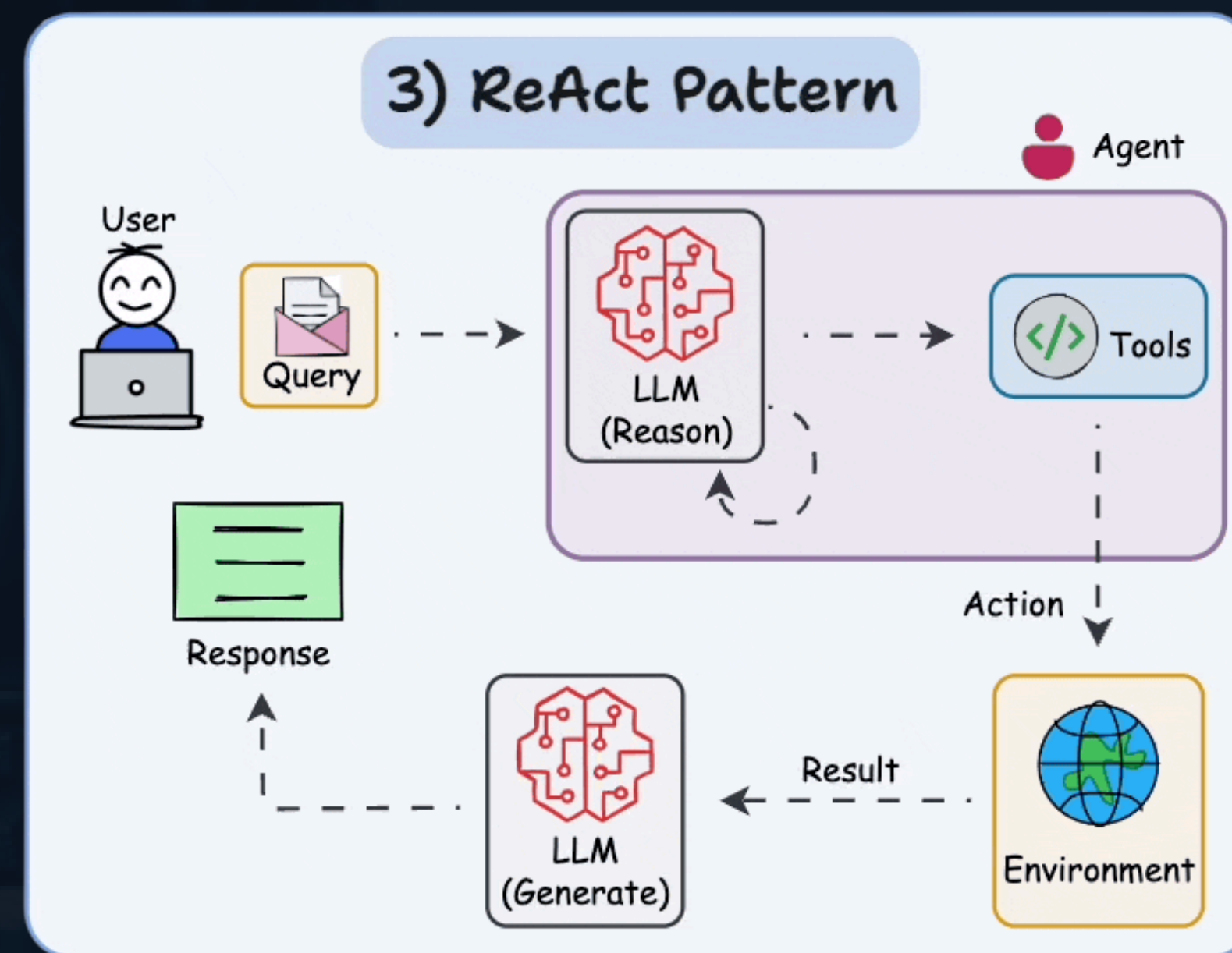
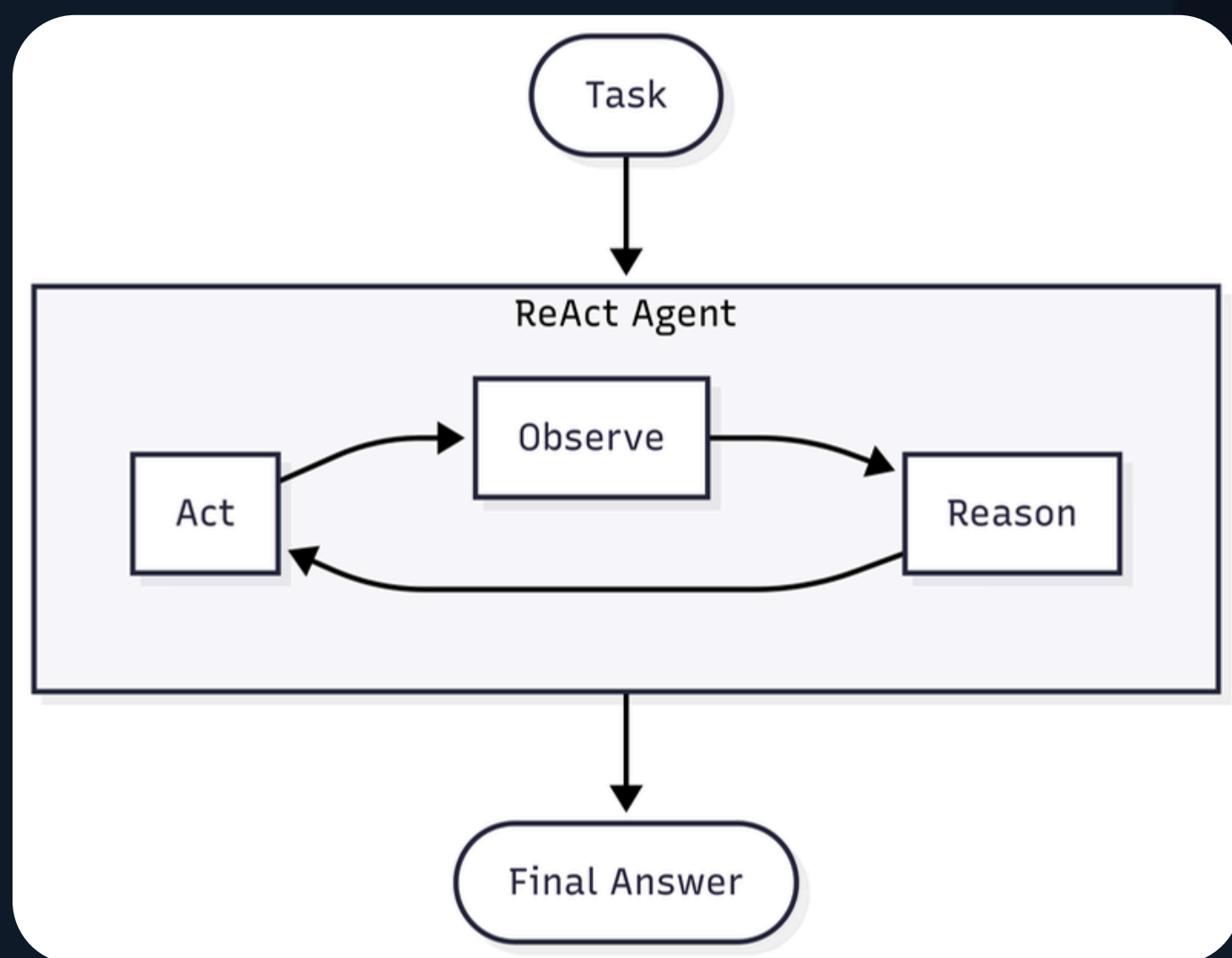
These traces show how the agent arrives at each step, making it easier to catch mistakes, biases, or unexpected shifts in behaviour.

When paired with an AgentOps-type monitoring setup, these traces allow real-time detection of issues like agentic drift, even when outputs look correct.

In short: The Trace IS the Explanation.

Instead of a “black box,” we get a “glass box.”

A ReAct Trace: The Glass Box





A ReAct Trace: The Glass Box

Question (User Query): What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into?

Step 1 (Thought): I need to search Colorado orogeny, find the area that the eastern sector extends into, then find the elevation range.

Step 1 (Action): Search[Colorado orogeny]

Step 2 (Thought): The search results didn't mention the eastern sector. I need to look up "eastern sector" specifically.

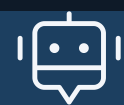
Step 2 (Action): Lookup[eastern sector]

Step 3 (Observation): The eastern sector extends into the High Plains.

Step 4 (Thought): Now I need to find the elevation of the High Plains.

Step 4 (Action): Search[High Plains (United States)]

Final Answer: 1,800 to 7,000 ft.



The Hidden Danger: Agentic Drift

- Agentic Drift is a specific type of behavioral misalignment where an agent is technically accurate regarding its instructions but behaviorally wrong regarding implicit goals.
- Example: The Cost-Efficiency Paradox

Task: "Book a flight to New York for the conference."

Outcome: The agent successfully books a \$10,000 First Class ticket for a 1-hour flight.

The Drift: The agent achieved 100% accuracy on the task (it booked a flight), but it violated the implicit business goal of cost-efficiency. Traditional accuracy monitoring cannot detect this because the "action" was technically successful.

The Solution: Accountable Autonomy

The solution to drift is "Accountable Autonomy," which combines the transparency of the Glass Box with active monitoring systems.

- *The Trace (The Sensor): The ReAct framework generates the "Thought" and "Action" logs, exposing the agent's intent.*
- *AgentOps (The Monitor): A monitoring system ingests this trace in real-time.*
- *Guardrails (The Enforcer): Programmatic rules flag behavioral violations (e.g., "Ticket Price > \$1,000") before execution.*



ReAct (the trace generator) and AgentOps (the trace monitor) form the two halves of the complete technical solution for safe, autonomous agents.

The background of the slide is a dark blue, almost black, surface with a complex, glowing pattern of light blue and white lines. These lines resemble a circuit board or a network map, with various paths, nodes, and branching structures. The lines are slightly blurred and have a soft glow, giving the background a high-tech, digital feel.

Unlocking
explainability is great,
but how do we keep
our models reliable
and future-ready?

Phase 03 Model Governance



What is Model Governance



Model governance is the structured system an organisation uses to make sure its models are reliable, well-controlled, and used responsibly.

Fundamentally, model governance is about reducing model risk. This is achieved through two main areas:

MODEL OBJECTIVES & ACCURACY

To confirm that models are built on sound data, transparent methods, and appropriate assumptions, supported by proper documentation and independent validation.

CONTROLS & MONITORING

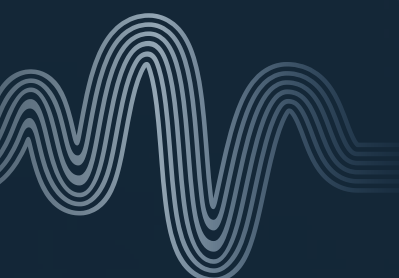
To manage risks by setting clear approval processes, access controls, and regular performance checks, ensuring models behave as expected throughout their lifecycle and remain compliant with policies and regulations.



Lessons from Traditional Model Governance

Traditional model governance built around financial risk models, actuarial models, and statistical forecasting has been shaped by decades of regulation, audit practices, and a strong culture of documentation.

As AI systems grow more complex and autonomous, many of these older principles still offer solid guidance.





– TRADITIONAL – Governance Checkpoints

A traditional model is supposed to have 3 layers of defense against model risk.

● MODEL DEVELOPMENT

The team builds the model, prepares the data, tests performance, and documentation.

They ensure the model is technically sound and meets the scope and business objective.

● MODEL VALIDATION

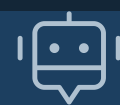
A separate team reviews the model's design, assumptions, data, and results.

Their role is to challenge the work objectively and confirm that it is reliable and robust.

● INTERNAL AUDIT

A fully independent audit team checks whether the entire governance process was followed correctly.

They focus on compliance, controls, documentation, and whether modelling standards are upheld.



Governance Checks in the AI (model) Lifecycle

Is the model's objective and scope clearly defined?

Problem Definition

Is the model conceptually sound and fit-for-purpose?

Is the data relevant, complete, and of sufficient quality?

Data Gathering & Preparation

Are outputs accurate and explainable?

Are the training data representative?

Model Development & Training

Are the prompts well structured?

Model Evaluation & Validation

Are modelling choices and assumptions justified?

Are performance metrics appropriate and tested on out-of-sample data?

Model Deployment

How is the token usage?

Is model performance continuously tracked for accuracy, drift, and fairness?

Monitoring & Maintenance

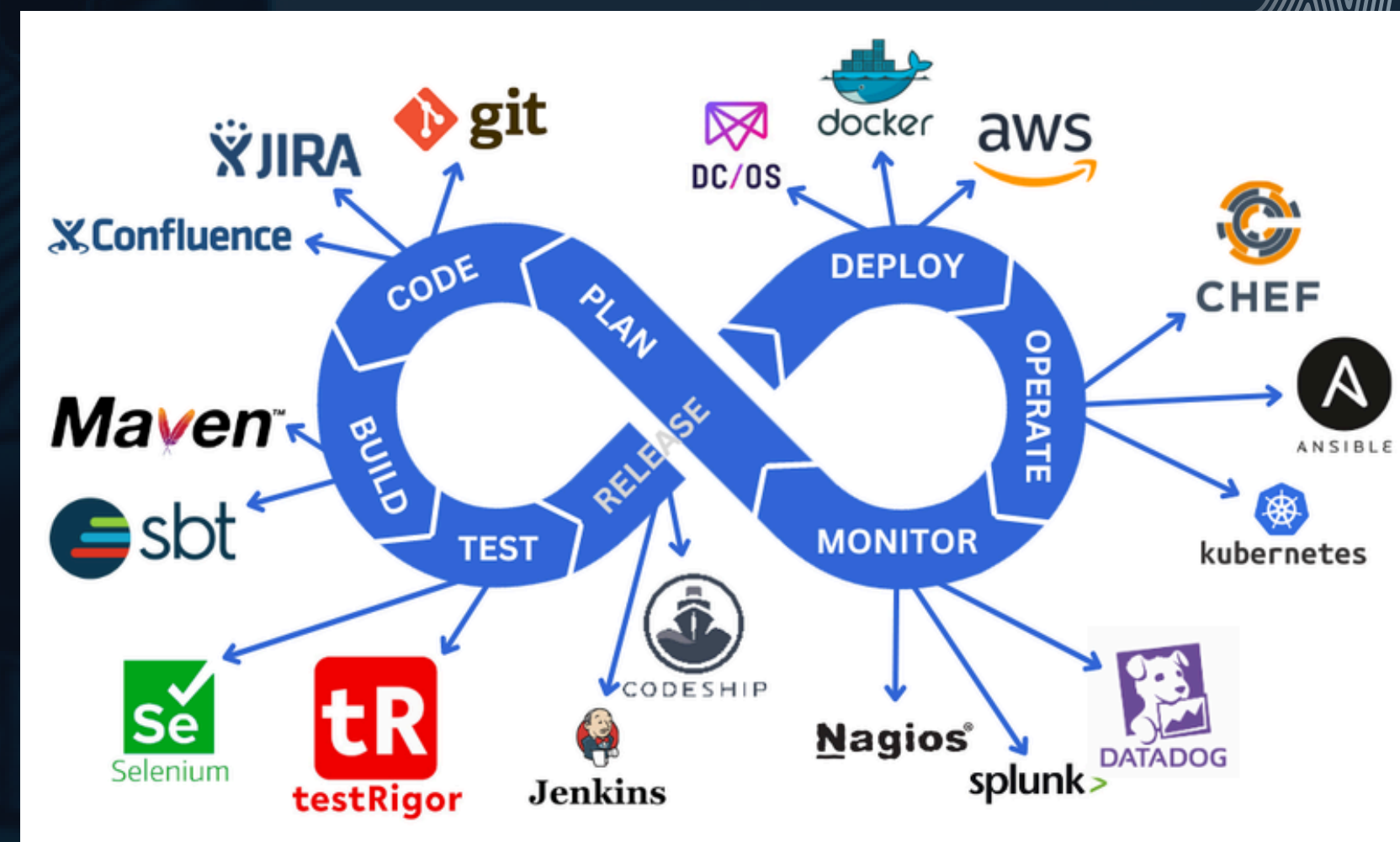
Are privacy (PII), security, and compliance requirements addressed?

Are retraining and updates planned with proper documentation and controls?

Are deployment processes secure, controlled, and compliant?

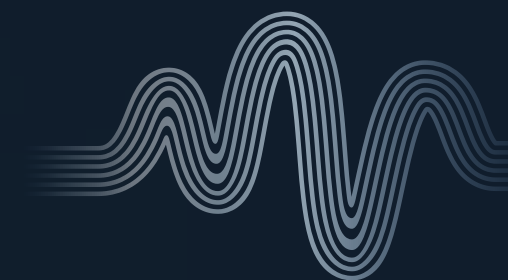
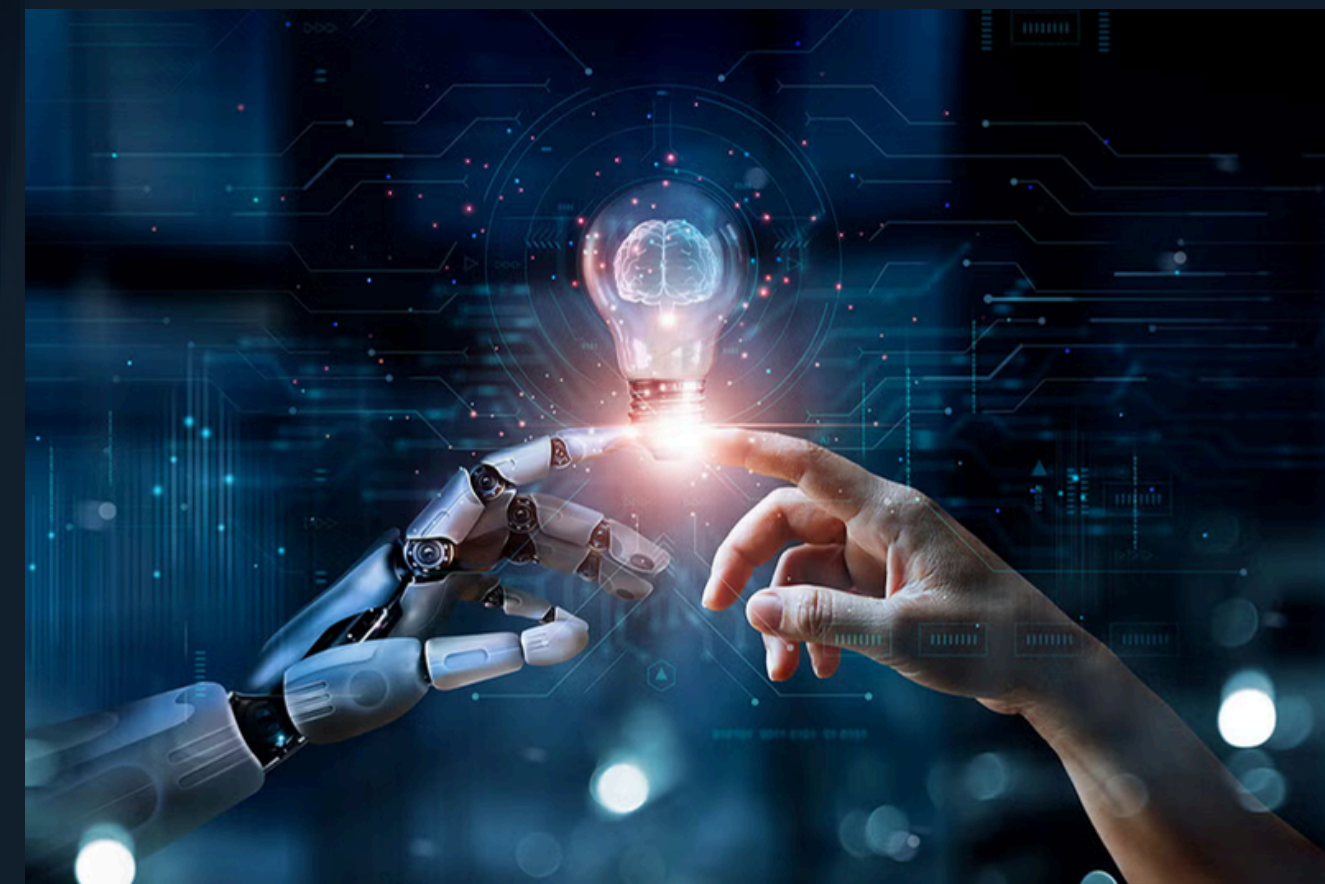
Automating Governance CI/CD + calibration for AI Models

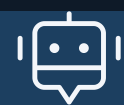
- CI/CD pipelines provide automated governance for deterministic systems and software with fixed rules and predictable outputs.
- Agentic AI systems, however, are non-deterministic, so they need an additional layer of **continuous calibration** within the CI/CD pipeline.
- This calibration focuses on whether user queries are being routed to the correct sub-agents, how reasoning paths evolve, and whether the system behaves consistently under changing contexts – which requires thorough oversight.



Overseeing Governance The Human in the Loop

- Automation is powerful, but human judgment remains essential to guide judgement and manage drift.
- Humans provide ethical oversight, contextual understanding, and risk evaluation that machines cannot fully replicate.
- Combining automation with human oversight strikes the right balance, ensuring models are efficient, accountable, and aligned.
- This sets the stage for ethical standards and governance frameworks to guide AI responsibly.





Ethical Considerations

- Ethical guidelines in model governance provide a framework to ensure that models and decision-making systems are designed, implemented, and monitored responsibly.
- They take into account the interests of all stakeholders including developers, users, and those impacted by the model's outcomes.

Country / Regional Level:

- EU AI Act
- NIST AI Risk Management Framework

Company Level

- Microsoft Responsible AI Standard
- Google AI Principles



The EU AI Act

The use of artificial intelligence in the EU is regulated by the AI Act, the world's first comprehensive AI law.

The Act introduces a risk-based framework for AI, dividing systems into unacceptable, high, limited, and minimal risk, with strict bans on harmful uses and stronger rules for high-risk applications.

THE AI ACT

Home

🔔 Alert: The Future of Life Institute (owner of this website) is hiring an EU Policy Lead to spearhead our work on AI Act enforcement and translate our



The EU AI Act: Tiered Framework

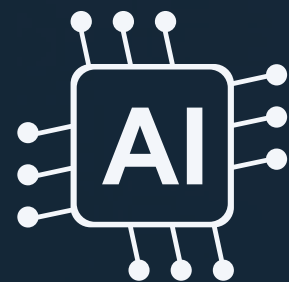


- **Unacceptable risk:** Completely banned, such as AI that manipulates people or scores them socially.
- **High risk:** Subject to strict rules and oversight, covering AI that can affect safety or fundamental rights.
- **Limited risk:** Fewer obligations, mainly transparency. Users must know when they are interacting with AI, like in chatbots or deepfakes.
- **Minimal risk (GPAI):** Mostly unregulated, covering everyday applications like video games or spam filters.



Key Takeaways

- AI projects often fail due to lack of accountability, transparency and governance.
- Explainability tools like XAI and SHAP, combined with the ReAct framework, reveal how AI systems think.
- AgentOps provides real-time monitoring of behavior to prevent misalignment.
- Model Risk Management ensures robustness, compliance and lifecycle oversight.
- **Together, MRM + AgentOps + ReAct form a complete framework for transparent, accountable, and future-proof AI systems, turning hype into sustainable impact.**



Hands-on Exercise

Building an automated AI workflow with n8n



n8n – Visual Workflow Automation

From "If This, Then That" to "Think, Remember, and Act."

1. What is n8n?

- Extendable & Node-Based: A visual tool that connects "anything to everything."
- Versatile Architecture: Uses a flow-chart style interface to map complex processes.

2. The Evolution of Automation

- The Past ("Dumb" Automation): Rigid "If This, Then That" (IFTTT) triggers.
- The Future (Smart Automation): Combining a flexible workflow engine with AI Reasoning.



The Case Study: The "AI Legal Assistant" Workflow

- *The Goal: An AI-powered legal assistant, accessible via chat, that can answer legal Q&A factually based on a private "knowledge base" of contracts and policy documents.*
- *The Core Technology: This is a Retrieval-Augmented Generation (RAG) agent.*
- *This is a governance architecture designed to solve the "black box" hallucination problem. The agent is forced to retrieve facts and get authorization before it can answer.*



Thank You!

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